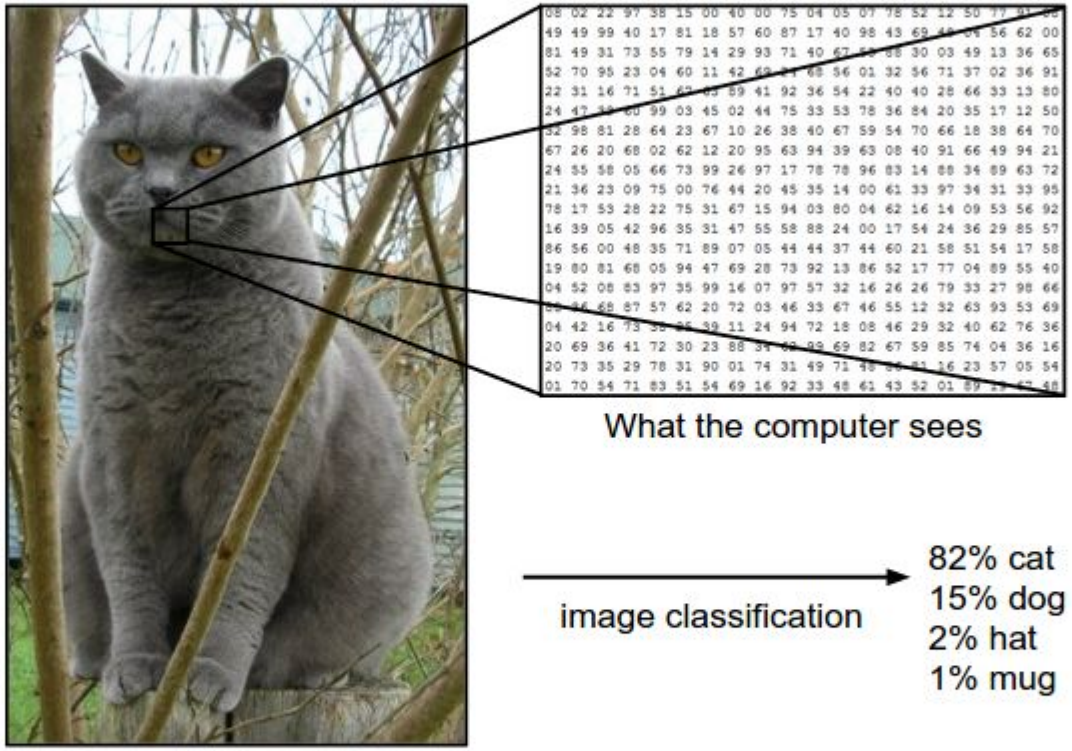
# CNN with Pytorch

## Mini Hackathon : 10 Marks

# Image Classification

- What we see and what machine see
- Pixel values of each segment of an image



# Problem Statement

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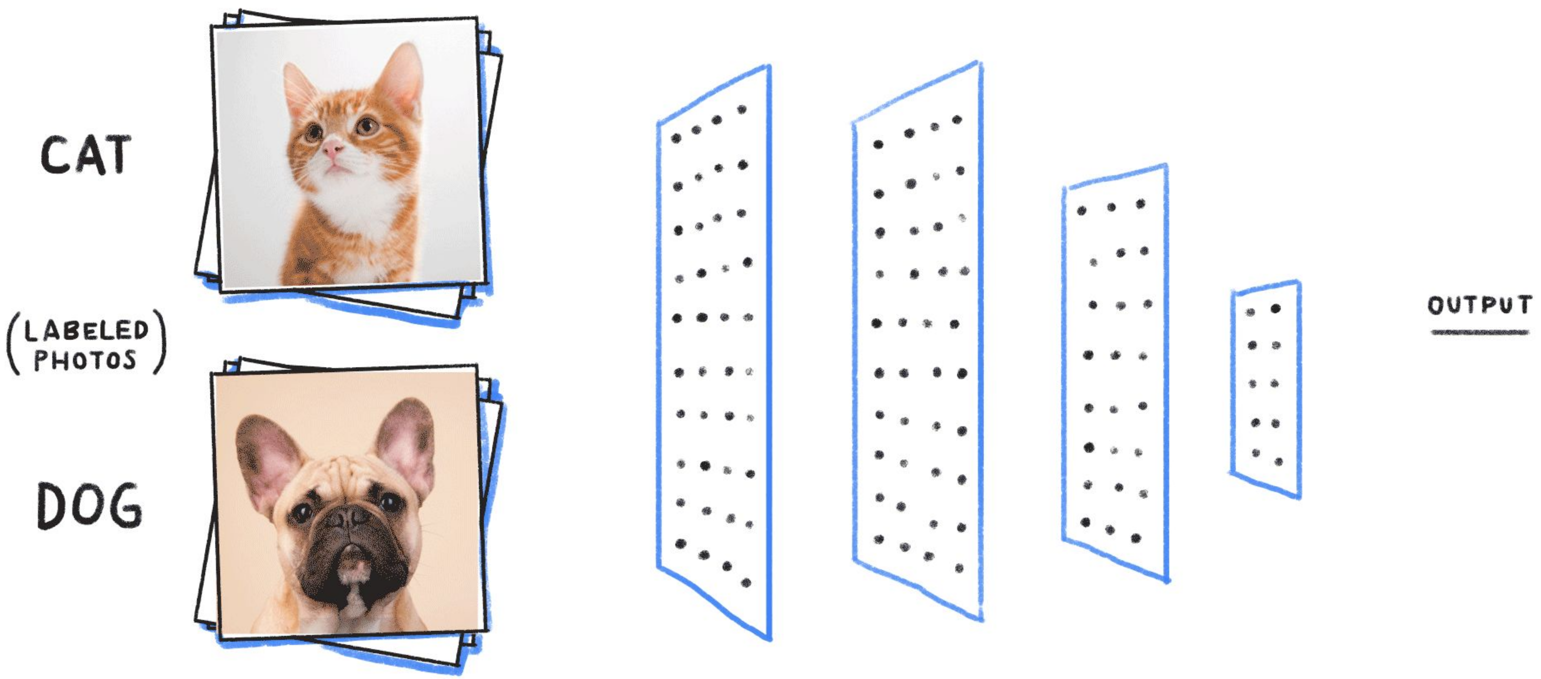
- To identify and classify images as cats or dogs



# Dataset & Learning Objectives

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- Dataset contains cats and dogs images
  - Train images ~22K including both cat and dog
  - Test images ~2K including both cat and dog

- In this mini hackathon
  - Load and prepare images
  - Develop a CNN model
  - Improve model performance

# Tasks

- **Stage 1:** Define Transformations and load data (3 Marks)

- **Stage 2**: Implement the CNN Model (4 Marks)
  - Sample Convolutional Layer and Maxpool Layer is provided

- **Stage 3:** Train the Model and validate it continuously to calculate the loss and accuracy for the train dataset across each epoch. (2 Marks)

- **Stage 4:** Testing Evaluation for CNN model (1 Marks)

# Evaluation

- Complete all the stages
- Expected training accuracy is above 90%
- Expected performance of test evaluation is above 90%
- Evaluation can be done on day of Mini Hackathon
- Timings **02:00 PM - 06:00 PM**
- MH1 scheduled - **Aug 01, 2026(Saturday)**

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# Thanks!!

## Questions ?